DAX MEASURES DOCUMENTATION

Model: Global Socio-Economic & Health Analytics **Author:** Amr Ramadan | Power BI Data Model

1. CORE FINANCIAL & POPULATION AGGREGATIONS

Essential numeric aggregations serving as base building blocks for core KPI cards across all dashboard pages.

Measure Name	DAX Formula Syntax	Return Type	Description
# Countries	`DISTINCTCOUNT('Country Data Comparison'[Country])`	Integer	Calculates unique count of countries/territories in filter context.
Total GDP	`SUM('Country Data Comparison'[GDP])`	Currency / Decimal	Aggregates Gross Domestic Product across all filtered regions.
AVG GDP	`AVERAGE('Country Data Comparison'[GDP])`	Currency / Decimal	Computes mean Gross Domestic Product per nation.
Total Population	`SUM('Country Data Comparison'[Population])`	Integer	Sums total global population count.
Total Exports	`SUM('Country Data Comparison'[Exports])`	Currency / Decimal	Aggregates total monetary exports value.
Total Imports	`SUM('Country Data Comparison'[Imports])`	Currency / Decimal	Aggregates total monetary imports value.
Total External Debts	`SUM('Country Data Comparison'[External Debt])`	Currency / Decimal	Sums total foreign debt obligations across nations.

2. ADVANCED LOGIC & DYNAMIC RANKING

GDP Rank

INTEGER (RANKING)

Ranks each country dynamically based on Total GDP in descending order using ALL to ignore row filters.

```
GDP Rank =
RANKX(
    ALL('Country Data Comparison'),
    [Total GDP],
    ,
    DESC,
    Skip
)
```

Highest GDP Country

TEXT / STRING

Dynamically identifies and returns the name of the top-performing economy ranked by GDP.

```
Highest GDP Country =
CALCULATE(
    SELECTEDVALUE('Country Data Comparison'[Country]),
    TOPN(1, 'Country Data Comparison', 'Country Data Comparison'[GDP], DESC)
```

3. DEMOGRAPHICS & HEALTH AVERAGES

Health and population growth averages calculated across selected geographical regions.

Measure Name	DAX Formula Syntax	Return Type	Description
AVG Life Expectancy	`AVERAGE('Country Data Comparison'[Life Expectancy at Birth])`	Decimal (Years)	Calculates mean life expectancy at birth across filtered countries.
AVG Birth Rate	`AVERAGE('Country Data Comparison'[Birth Rate Per 1k])`	Decimal (per 1,000)	Computes average annual birth rate per 1,000 population.
AVG Death Rate	`AVERAGE('Country Data Comparison'[Death Rate Per 1k])`	Decimal (per 1,000)	Computes average annual mortality rate per 1,000 population.

4. DERIVED & ANALYTICAL RATIO METRICS

Population Density Indicator

DECIMAL (RATIO)

Calculates population density per square kilometer using safe divide against total land area.

```
Population Density Indicator =
VAR Total_Area = SUM('Country Data Comparison'[Area SQKm])
RETURN
    DIVIDE([Total Population], Total_Area, 0)
```

Trade Balance Indicator

TEXT / CATEGORICAL

Evaluates macroeconomic net trade balance (Exports minus Imports) and assigns dynamic trade status labels.

```
Trade Balance Indicator =
VAR Total_Imports = SUM('Country Data Comparison'[Imports])
VAR Total_Exports = SUM('Country Data Comparison'[Exports])
VAR Net_Trade = Total_Exports - Total_Imports
RETURN
    IF(
        ISBLANK(Total_Exports) || ISBLANK(Total_Imports), "No Data",
        IF(Net_Trade > 0, "Trade Surplus", "Trade Deficit")
    )
```